Information Rates Over Multi-View Channels

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***Abstract*—In this paper, we investigate the fundamental limits of reliable communication over communication channels when there are a large number of noisy views of a transmitted symbol, in other words, in the asymptotic regime when several copies of a single symbol are sent independently through the channel. We consider first the setting of *multi-view* discrete memoryless channels (DMCs) and then extend our results to general multi-view multi-letter channels. We argue that the channel capacity and dispersion of such multi-view channels converge exponentially quickly in the number of views to to the entropy and varentropy of the input distribution, respectively, and identify the exact rate of convergence. This rate is governed by the smallest Chernoff information between two conditional distributions of the output given unequal inputs. For the special case of the deletion channel, we compute upper bounds on this Chernoff information and discuss some consequences. Finally, we present a new channel model that we call the *Poisson approximation channel*—of possible independent interest—whose capacity closely approximates the capacity of the multi-view binary symmetric channel (BSC) for any *fixed* number of views.**

I. Introduction

Consider a communication setting where the decoder receives multiple, potentially noisy, views of a single transmitted sequence that is sent via independent transmissions through a noisy channel. Such a scenario arises naturally, for example, in DNA-based storage systems (see, e.g., [1]–[4] and [5], [6] for coding schemes). In such systems, the (short) sequenced DNA molecules are "amplified" by Polymerase Chain Reaction (PCR), to produce several copies of each molecule. The resultant pool of molecules is then sampled from, and the sampled molecules subjected to noisy reads, during sequencing. Clearly, multiple noisy views (or multiple draws) of a transmitted sequence can only improve the error probabilities of decoding, as compared to decoding a single received sequence. Viewed differently, it is clear that the Shannon capacity of the *multi-view* channel, as described above, is at least as large as the capacity of the standard single-view channel.

In this paper, we consider first the setting where the noisy channel is a discrete memoryless channel (DMC) and then extend our results to general multi-letter channels. Multi-view DMCs were first studied in the classic work by Levenshtein [7] on sequence reconstruction. Among several other results, [7] presented bounds on the probability of error under ML decoding and characterized the number of noisy views required to guarantee recovery of the transmitted sequence, with *decaying error probability*. These results are determined by the minimum Chernoff information between conditional distributions of outputs of the DMC, given unequal inputs. Levenshtein's result on the number of noisy views required for reconstruction, however, depends crucially on the assumption that the error probability decays with the blocklength, and leaves open the question of characterizing the rates achievable over DMCs with fixed error probability, using finitely many channel uses. Furthermore, the proof strategy in [7] resists extension to channels, such as those that model synchronization errors, wherein the lengths of the input and output sequences could be different.

In this work, we focus on coded communications and exactly characterize the rate of convergence of the information rate and the channel dispersion of general multi-view channels, under arbitrary input distributions, to the input entropy and the varentropy, respectively, when the number of views is large. We show that our results also hold for multi-letter channels with potentially different lengths of input and output sequences. As our main result, we show that the rates of (exponential) convergence are again determined by the minimum Chernoff information between conditional distributions, as above. An important consequence of our results for DMCs (under some regularity conditions) is our characterization of the largest rates achievable over finitely many channel uses, with any *fixed error probability*, using results from finite-blocklength information theory, thereby resolving a natural question that stems from Levenshtein's work [7]. Furthermore, for the special case of the (multi-letter) deletion channel, which is a popular synchronization error channel, we compute explicit upper bounds on the above Chernoff information, and discuss some interesting consequences.

For DMCs, our results, which do not assume any additional structure such as symmetry, or size of the input and output alphabets (besides finiteness), are significant in the context of DNA-based storage, where the inputs are drawn from a quarternary alphabet and the DMC that models sequencing errors is asymmetric [8]. For DNA-based storage channels, for example, following [4, Thm. 5], our results help provide an upper bound on the number of times the stored DNA molecules must be sampled, as a function of their length, so that the resulting channel (arising due to sampling and sequencing) is almost noiseless.

While the results discussed above consider the asymptotic regime of large numbers of views, another interesting question that can be asked is the characterization of the capacities of special multi-view DMCs, for a fixed number of views. In this non-asymptotic setting, the capacities of multi-view DMCs were studied in [9], wherein the capacity of the multi-view binary symmetric channel (BSC) was identified and an explicit expression was provided. More generally, bounds on the capacities of multi-view DMCs were derived in the literature on "information combining" (see [10], [11] and the references therein). Also

related is the well-studied problem on trace reconstruction [12]–[14], wherein an input string is passed independently through a deletion channel and the goal is to find the number of resultant outputs required for reconstruction with high-probability.

In this work, we also explore the non-asymptotic capacity of the so-called *Poisson approximation channel*, which we believe is of possible independent interest. We show that the capacity of this channel is a tight lower bound on the capacity of the multi-view BSC.

The paper is organized as follows: Section II sets down the notation; Section III presents our main results for DMCs; Section IV provides a proof of the rate of convergence of the information rate of a DMC to the input entropy and Section V provides a proof of the rate of convergence of the channel dispersion of a DMC to the input varentropy. Section VI then extends our results to general multi-letter channels and presents upper bounds on Chernoff information discussed above for the deletion channel. Section VII contains a proof that the capacity of the Poisson approximation channel is a tight lower bound on the capacity of the multi-view BSC. Section VIII concludes the paper and proposes problems for further research.

II. Notation

All logarithms are assumed to be natural logarithms. Random variables are denoted by capital letters, e.g., X, Y, and small letters, e.g., x, y, denote their instantiations. Sets are denoted by calligraphic letters, e.g., $\mathcal{X}, \mathcal{Y}$. The set of positive natural numbers $\{1, 2, \ldots\}$ is denoted as $\mathbb{N}$. Notation such as $P(x), P(y|x)$ are used to denote the probabilities $P_X(x), P_{Y|X}(y|x)$, when it is clear which random variables are being referred to. The notation $\mathsf{H}(X) := \mathbb{E}[-\log P(X)]$ and $\mathsf{V}(X) := \mathbb{E}\left[(-\log P(X) - \mathsf{H}(X))^2\right]$ denote the entropy and varentropy of X, which depend only on the distribution of X. The Chernoff information $\mathsf{C}(P, Q)$ between two distributions P and Q defined on the same alphabet $\mathcal{X}$ is given by (see [15, Ch. 11])

$$\mathsf{C}(P, Q) := -\min_{\lambda \in [0,1]} \log\left(\sum_{x \in \mathcal{X}} P(x)^{1-\lambda} Q(x)^{\lambda}\right). \tag{1}$$

Further, the standard Kullback-Leibler (KL) divergence between two distributions P, Q on the same alphabet $\mathcal{X}$ is given by $D(P||Q) := \mathbb{E}_{X\sim P}[\log \frac{P(X)}{Q(X)}]$, where we take $0 \log \frac{0}{z} = 0$ and we set $D(P||Q) = \infty$ if there exists $x \in \mathcal{X}$ such that $Q(x) = 0$ but $P(x) > 0$. The notation a^n is used to denote the length-n vector $(a_1, \ldots, a_n)$, and we define $w(a^n)$ to be the Hamming weight, or the number of ones in a^n.

For a symbol b, the notation $\mathbb{1}\{x = b\}$ is used to denote the indicator that x equals b; this indicator equals 1 if $x = b$ and 0, otherwise. For sequences $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ of positive reals, $a_n = o(b_n)$ if $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$, that $a_n = O(b_n)$ if for all n large enough, $a_n \leq c \cdot b_n$, for some $c > 0$, and that $a_n = \Theta(b_n)$ if for all n large enough, we have that $m \cdot b_n \leq a_n \leq M \cdot b_n$, for some constants $m, M \in \mathbb{R}$.

III. Main Results for DMCs

We now formalize the problem for the case when the noisy channel is a DMC and present our main results. Section VI then extends Theorem III.1 here to general multi-letter channels. Consider a DMC W with finite input and output alphabets $\mathcal{X}$ and $\mathcal{Y}$, respectively, specified by the channel law $\{P_{Y|X}(y|x) = W(y|x) : \ x \in \mathcal{X}, y \in \mathcal{Y}\}$. Let $X \sim P_X$ be the input to the channel. Our setting of interest is when $d \geq 1$ noisy outputs are obtained by independently passing the input through the DMC. Let these outputs be denoted by $Y_1, \ldots, Y_d$; their conditional distribution obeys

$$P(Y^d = y^d \mid X = x) = \prod_{i=1}^{d} W(y_i|x_i).$$

We call a DMC with a channel law as above as the *d-view* DMC $W^{(d)}$. We assume further that $|\mathcal{X}|$ and $|\mathcal{Y}|$ do not depend on d.

Now, given any DMC W, we denote its capacity by $C(W) = \max_{P_X} I(X;Y) = \max_{P_X} \mathbb{E}\left[\log \frac{W(Y|X)}{\sum_{x\in\mathcal{X}} P(x)W(Y|x)}\right]$ [16]. Hence, the capacity of $W^{(d)}$ is given by $C^{(d)} := \max_{P_X} I^{(d)}$, where $I^{(d)} := I(X;Y^d)$. Further, for any input distribution P_X, the channel dispersion of the d-view DMC W is given by

$$V^{(d)} := \mathbb{E}\left[\left(\iota(X;Y^d) - I(X;Y^d)\right)^2\right],$$

where $\iota(x;y^d) := \log \frac{P(y^d|x)}{P(y^d)}$. We are interested in characterizing the behaviour of $C^{(d)}$ (or more generally, of $I^{(d)}$) and $V^{(d)}$ when d is large. From operational considerations, we can argue that $I^{(d)}$ and $V^{(d)}$ are non-decreasing in d (more formally, for $I^{(d)}$, the data processing inequality [15, Thm. 2.8.1] guarantees that $I^{(d)}$ is non-decreasing in d). Furthermore, it is reasonable to expect that $I^{(d)}$ and $V^{(d)}$ converge to $\mathsf{H}(X)$ and $\mathsf{V}(X)$, respectively, since for very large d, the "uncertainty" introduced by the channel is expected to drop to 0. In this work, we confirm these convergence results by showing that the $I^{(d)}$ and $V^{(d)}$ in fact converge to their limiting values exponentially quickly in d, and explicitly identify the rate of convergence.

To build some intuition why the convergence is exponentially fast in d, at least for $I^{(d)}$, consider the simple example of the d-view BSC(p), which we denote by BSC$^{(d)}(p)$ (here, $p < 1/2$ is the crossover probability of the BSC), with a uniform input distribution. At the decoder end, we construct a simple "data processor" that takes the (random) outputs $Y^d \in \{0,1\}^d$ and computes their majority $M \in \{0,1\}$. Then,

$$\begin{aligned} I^{(d)} &= I(X;Y^d) \\ &\geq I(X;M) = 1 - h_b(\Pr[X \neq M]), \end{aligned}$$

where the inequality holds by the data processing inequality and the last equality holds by symmetry. Here, h_b is the binary entropy function, with $h_b(\gamma) := -\gamma \log \gamma - (1-\gamma)\log(1-\gamma)$, for $\gamma \in [0,1]$. Now, observe that since $p < 1/2$, we have by the Chernoff bound (see [17, Example 1.6.2]) that

$$\begin{aligned} \Pr[X \neq M] &\leq \exp(-d \cdot D(\mathrm{Ber}(1/2)||\mathrm{Ber}(p))) \\ &= \exp(-d \cdot Z(p)), \end{aligned}$$

where $Z(p) = 2\sqrt{p(1-p)}$ is the Bhattacharya parameter for the BSC(p). Hence, since $h_b(\gamma) \leq -\gamma \log \gamma$, for all $\gamma \in [0,1]$, we see that $I^{(d)} \geq 1 - dZ(p) \cdot \exp(-d \cdot Z(p))$, thereby providing reason for the intuition that the convergence of $I^{(d)}$ to 1 is

exponentially fast in d (in fact, in Theorem III.1, we argue that the rate $Z(p)$ is in fact tight for the BSC(p)).

Our main theorem is given below.

Theorem III.1. *We have that*

$$I^{(d)} = H(X) - \exp\left(-d\rho + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right), \text{ and}$$
$$V^{(d)} = \exp\left(-d\rho + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right),$$

where $\rho = \min_{x,x':x\neq x'} \mathsf{C}(P_{Y|x}, P_{Y|x'})$.

Note that the theorem above captures the dependence of the speed of convergence of mutual information and dispersion to their limits in terms of both the number of views d and the size of the input alphabet (assumed to be independent of d). The dependence on $|\mathcal{X}|$ will assume significance when we transition to multi-letter channels in Section VI.

We now proceed to interpret Theorem III.1. Suppose we define the rates

$$\rho(I) := \liminf_d -\frac{1}{d}\cdot\log\left(\mathsf{H}(X) - I^{(d)}\right)$$
$$= \liminf_d -\frac{1}{d}\cdot\log\mathsf{H}(X|Y^d), \text{ and}$$
$$\rho(V) := \liminf_d -\frac{1}{d}\cdot\log\left(\mathsf{V}(X) - V^{(d)}\right),$$

we get by the above result that these rates of convergence obey $\rho(I) = \rho(V) = \rho$, which in turn is independent of the input distribution P_X, with possible dependence on its support. Further, the limits in the definitions of these rates exist. Further, for a binary-input memoryless symmetric (BIMS) channel W (see [18, Chap. 4] for a definition), we have that the function $f(\lambda) := \log\left(\sum_{y\in\mathcal{Y}} P(y|0)^{1-\lambda}P(y|1)^{\lambda}\right)$ is symmetric about $\lambda = 1/2$. Hence, using the fact that f is convex, we obtain that the minimum in (1) is attained at $\lambda = 1/2$, giving

$$\mathsf{C}(P_{Y|0}, P_{Y|1}) = -\log\sum_{z\in\mathcal{Z}}\sqrt{P_{Y|0}(z|0)P_{Y|1}(z|1)}$$
$$= -\log Z_b(W),$$

where $Z_b(W)$ is the Bhattacharya parameter of the BIMS channel W. Hence, following the results in [19], the rates of convergence for a BIMS channel W are extremized by a BEC and BSC having the same capacity as W, respectively, with the BEC having the largest and BSC having the smallest Bhattacharya parameters, respectively. Here, for the d-view BSC(p), we have $\rho(I) = \rho(V) = -\log 2\sqrt{p(1-p)}$, and for the d-view BEC(ϵ), we have $\rho(I) = \rho(V) = -\log\epsilon$. Furthermore, for the Z-channel with parameter δ, $\rho(I) = \rho(V) = -\log\delta$. In fact, for most well-behaved DMCs W, we will have that $\rho(I) = \rho(V) > 0$, implying exponential convergence of mutual information and dispersion to $\mathsf{H}(X)$ and $\mathsf{V}(X)$, respectively.

Theorem III.1 along with the normal approximation for finite blocklengths [20] then allow us to characterize, up to exponential tightness, the largest finite-blocklength rates achievable over a d-view DMC $W^{(d)}$, under some regularity conditions. Our presentation follows that in [21] (see also [22, Thm. 22.2]). Suppose that the DMC $W^{(d)}$ is non-singular (see the definition in [21, Sec. 4.2.1]). Let $M^\star(n,\epsilon)$ be the largest integer M such that there exists a blocklength-n code with maximal probability of error $\epsilon > 0$ over $W^{(d)}$. Further, let $P_X^\star$ be that capacity-achieving input distribution for $W^{(d)}$ that minimizes $V^{(d)}$; we assume that $P_X^\star$ is independent of d and that under this distribution, $V^{(d)} > 0$. For $t \in \mathbb{R}$, let $Q(t) = \int_t^\infty \frac{1}{\sqrt{2\pi}}e^{-z^2/2}\mathrm{d}z$, with Q^{-1} denoting its inverse. We then have the following theorem:

Theorem III.2. *We have that for any non-singular DMC $W^{(d)}$, for all $n \geq 1$ and $\epsilon \in (0,1)$, under the capacity-achieving distribution $P_X^\star$,*

$$\frac{\log M^\star(n,\epsilon)}{n}$$
$$= H(X) - e^{-d\rho+\Theta(\log d)} + Q^{-1}(\epsilon)\cdot\frac{e^{-d\rho/2+\Theta(\log d)}}{\sqrt{n}} + O\left(\frac{\log n}{n}\right),$$

where $\rho = \rho(I) = \rho(V)$.

Clearly, it follows from the above theorem that for any finite n, by choosing $d = \rho^{-1}\log n$, we can obtain rates up to $\frac{\log M^\star(n,\epsilon)}{n} = H(X) - O\left(\frac{\log n}{n}\right)$ over any d-view DMC $W^{(d)}$.

Next, we consider the special case of the d-view BSC(p), which we denote by BSC$^{(d)}(p)$; the capacity of this channel is known to be equal to the capacity of the so-called binomial channel [9, Sec. III.A], denoted by $\mathsf{Bin}_d(p)$. This capacity is given by

$$C(\mathsf{Bin}_d(p))$$
$$= 1 + \sum_{i=0}^{d}\binom{d}{i}\left(p^i(1-p)^{d-i}\log\frac{p^i(1-p)^{d-i}}{p^i(1-p)^{d-i} + p^{d-i}(1-p)^i}\right), \tag{2}$$

which is achieved when the input distribution $P_X = \mathrm{Ber}(1/2)$. While Theorem III.1 determines the rate of convergence of $C\left(\mathrm{BSC}^{(d)}(p)\right) = C(\mathsf{Bin}_d(p))$ (via the rate of convergence of the mutual information $I(X;Y^d)$ under a uniform input distribution $P_X = \mathrm{Ber}(1/2)$) to $H(P_X) = 1$ in the asymptotic regime of large d, it will be useful if a more fine-grained, non-asymptotic understanding of the capacity of this multi-view channel is obtainable. To this end, we identify a closely-related DMC, which we call the *Poisson approximation channel* and denote by $\mathsf{Poi}_d(p)$, whose capacity turns out to be a good approximation to $C(\mathsf{Bin}_d(p))$. The DMC $\mathsf{Poi}_d(p)$ has input alphabet $\mathcal{X} = \{0,1\}$ and output alphabet $\mathcal{R} = (\mathbb{N}\cup\{0\})\times(\mathbb{N}\cup\{0\})$. We denote the input bit as X and the output as the pair (R_1, R_2). The channel law obeys

$$P_{R_1,R_2|X}(r_1,r_2|x) = P_{R_1|X}(r_1|x)P_{R_2|x}(r_2|x),$$
$$\text{for all } x\in\{0,1\} \text{ and } (r_1,r_2)\in(\mathbb{N}\cup\{0\})\times(\mathbb{N}\cup\{0\}),$$

where

$$P_{R_1|0} = \mathrm{Poi}(d(1-p)),\ P_{R_2|0} = \mathrm{Poi}(dp), \text{ and}$$
$$P_{R_1|1} = \mathrm{Poi}(dp),\ P_{R_2|1} = \mathrm{Poi}(d(1-p)).$$

Here, $\mathrm{Poi}(\gamma)$ denotes the Poisson distribution with parameter γ. Note that $\mathsf{Poi}_d(p)$ is a BIMS channel; hence, its capacity is achieved by $P_X = \mathrm{Ber}(1/2)$ (see [18, Problem 4.8]). The following theorem then holds.

Theorem III.3. *We have that*

$$C(\mathsf{Poi}_d(p)) \le C(\mathsf{Bin}_d(p)) \le C(\mathsf{Poi}_d(p)) + \exp(-d(1-Z(p))) - Z(p)^{2d},$$

where $Z(p) = 2\sqrt{p(1-p)}$ is the Bhattacharya parameter of the BSC(p).

Figure 1 plots the capacities of the $\mathsf{Bin}(p)$ and $\mathsf{Poi}(p)$ channels when $d = 24$. It is clear from the plots and from Theorem III.3 that $C(\mathsf{Poi}_d(p))$ is a tight lower bound on $C(\mathsf{Bin}_d(p))$.

We now discuss a simple extension of Theorem III.3 to d-view BIMS channels $W^{(d)}$ with a finite output alphabet. It is well-known (see, e.g., Section II in [23]) that any BIMS channel W with a finite output alphabet can be decomposed into finitely many BSC subchannels, i.e.,

$$W_{Y|X=x} = \sum_{i=1}^{K} \epsilon_i \cdot W^{(i)}_{Y|X=x},$$

for some $K < \infty$, for any $x \in \{0,1\}$, with $W^{(i)}_{Y|X=x}$ being the channel law of a BSC with crossover probability p_i. Here, ϵ_i is the probability of choosing subchannel i, with $\sum_i \epsilon_i = 1$. Let P_s be the distribution on $\{1,\dots,K\}$ with mass ϵ_i at point i. We then have that

$$C^{(d)}(W) = \mathbb{E}\left[C(J_1,\dots,J_d)\right], \tag{3}$$

where the random variables J_i, $1 \le i \le K$, are drawn i.i.d. according to P_s, and the notation $C(j_1,\dots,j_d)$ refers to the capacity of that channel with the channel law

$$P(Y^d = y^d | X = x) = \prod_{\ell=1}^{d} W^{(j_\ell)}(y_\ell|x),$$

for $x \in \{0,1\}$ and $y^d \in \mathcal{Y}^d$. Now, suppose that $0 < p_1 \le p_2 \le \dots p_K < 1/2$. We then have from Theorem III.3 and (3) that

$$C(\mathsf{Poi}_d(p_K)) \le C^{(d)}(W) \le C(\mathsf{Poi}_d(p_1)) + \exp(-d(1-Z(p_1))) - Z(p_1)^{2d}.$$

IV. Convergence of Mutual Information of DMCs

In this section, we prove Theorem III.1 for the mutual information (or information rate) $I^{(d)}$ of a DMC W with an aribitrary distribution P_X.

A. Convergence of Mutual Information for BIMS Channels

To build intuition, we present a simple proof for the case when the DMC W is a BIMS channel, with $P_X = \mathrm{Ber}(1/2)$. The choice of the input distribution being uniform assumes relevance since the capacity of a BIMS channel is achieved by uniform inputs (see [18, Problem 4.8]).

The following bounds on the conditional entropy of the input to a BIMS channel given the output, which are drawn from [24, Prop. 2.8], will prove useful. Recall the definition of the Bhattacharya parameter of a BIMS channel W with output alphabet $\mathcal{Y}$ given by $Z_b(W) := \sum_{y\in\mathcal{Y}} \sqrt{P_{Y|0}(y)P_{Y|1}(y)}$.

Lemma IV.1. *Given a BIMS channel $\overline{W}$ with input $\overline{X} \sim P_X$, where $P_X = Ber(1/2)$, and output $\overline{Y}$, we have that*

$$Z_b(\overline{W})^2 \le \mathsf{H}(\overline{X} \mid \overline{Y}) \le \log(1 + Z_b(\overline{W})).$$

We thus obtain the following simple corollary.

Corollary IV.1. *Given a BIMS channel W with input $X \sim P_X$, where $P_X = Ber(1/2)$, we have that for the d-view DMC $W^{(d)}$,*

$$Z_b(W)^{2d} \le \mathsf{H}(X \mid Y^d) \le Z_b(W)^d.$$

Proof. We first note that when W is a BIMS channel, then so is the d-view DMC $W^{(d)}$. Now, the Bhattacharya parameter

$$\begin{aligned} Z_b(W^{(d)}) &= \sum_{y^d\in\mathcal{Y}^d} \sqrt{P_{Y|0}(y^d|0)P_{Y|1}(y^d|1)} \\ &= \sum_{y^d\in\mathcal{Y}^d} \prod_{i=1}^{d} \sqrt{P_{Y|0}(y_i|0)P_{Y|1}(y_i|1)} \\ &= \prod_{i=1}^{d} \sum_{y_i\in\mathcal{Y}} \sqrt{P_{Y|0}(y_i|0)P_{Y|1}(y_i|1)} \\ &= Z_b(W)^d. \end{aligned}$$

The proof is then completed by appealing to Lemma IV.1 and noting that $\log(1+z) \le z$, for all $z > -1$. □

Moreover, we have the following bound on $\rho(I)$ for general binary-input DMCs and for arbitrary input distributions P_X.

Lemma IV.2. *Given a binary-input DMC W with input $X \sim P_X$, where P_X is arbitrary, we have that*

$$\mathsf{H}(X|Y^d) \ge e^{-d\cdot\mathsf{C}(P_{Y|X=0},P_{Y|X=1})+\Theta(\log d)}.$$

Proof. Observe that

$$\mathsf{H}(X|Y^d) = \mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(X \mid Y^d)}\right].$$

Further, recall that in a binary Bayesian hypothesis testing setting (see [15, Ch. 11]) with prior P_X and hypotheses $\{P_{Y|0}, P_{Y|1}\}$, we have that the probability of correct decision is given by

$$\mathbb{E}\left[P(X \mid Y^d)\right] = 1 - e^{-d\cdot\mathsf{C}(P_{Y|X=0},P_{Y|X=1})+\Theta(\log d)}.$$

Hence, we see that for large enough d,

$$\begin{aligned} \mathsf{H}(X|Y^d) &= \mathbb{E}\left[\log \frac{1}{P(X \mid Y^d)}\right] \\ &\ge -\log\left(1 - e^{-d\cdot\mathsf{C}(P_{Y|X=0},P_{Y|X=1})+\Theta(\log d)}\right) \\ &\ge e^{-d\cdot\mathsf{C}(P_{Y|X=0},P_{Y|X=1})+\Theta(\log d)}, \end{aligned} \tag{4}$$

where the first inequality follows from Jensen's inequality and the second from the fact that $\log(1+z) \le z$, for $z > -1$. Also, the $\Theta(\log d)$ term in the exponent in (4) is due either to the Laplace method (see [25, Sec. 4.2]) or to the results of [26, Thm. 2.1 and Corollary 1].

□

The above ingredients suffice to afford a proof of Theorem III.1 for the special case of BIMS channels with a uniform input distribution.

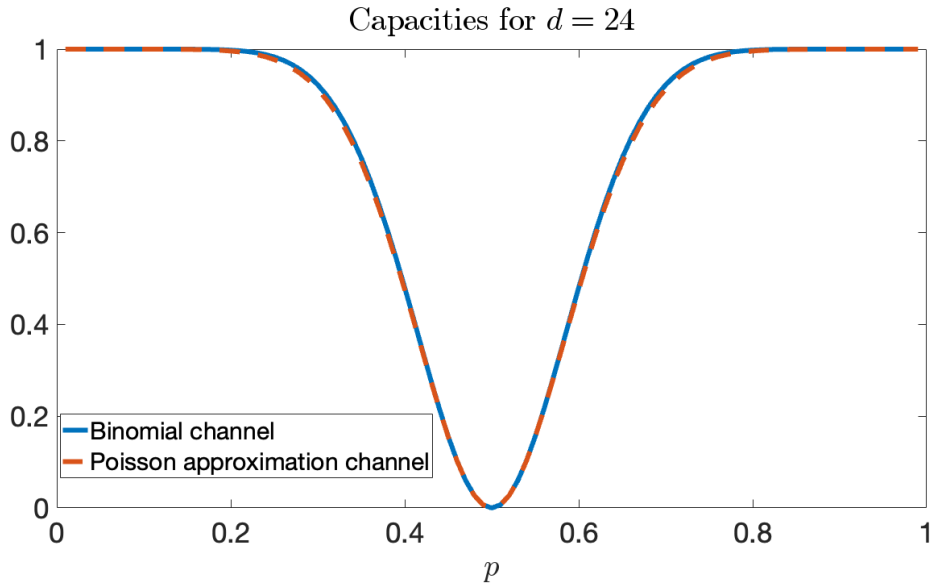

Fig. 1: Plots comparing $C(\mathsf{Bin}(p))$ and $C(\mathsf{Poi}(p))$ for $p \in [0, 1]$, when $d = 24$.

Proof of Thm. III.1 for BIMS channels with uniform inputs. For the case when W is a BIMS channel, we have that $\mathsf{C}(P_{Y|0}, P_{Y|1}) = -\log Z_b(W)$ (see the remark following the statement of Theorem III.1). Hence, when the input distribution is uniform, we have from Corollary IV.1, we have that $\mathsf{H}(X|Y^d) \leq e^{-d\cdot\mathsf{C}(P_{Y|0},P_{Y|1})}$. The proof concludes by appealing then to Lemma IV.2, which shows that $\mathsf{H}(X|Y^d) \geq e^{-d\cdot\mathsf{C}(P_{Y|0},P_{Y|1})+\Theta(\log d)}$. Hence, overall, we have that $\mathsf{H}(X|Y^d) = e^{-d\cdot\mathsf{C}(P_{Y|0},P_{Y|1})+\Theta(\log d)}$. □

It is possible, with some work, to present a proof of Theorem III.1 for $I^{(d)}$ for general DMCs with arbitrary input distributions with the aid of [7, Lemma 9] (see also [27, Thm. 5]) and an extension of Lemma IV.2. However, the analysis of dispersion requires more involved techniques: we hence first present a proof for $I^{(d)}$ that introduces these techniques, in Section IV-B, and then extend them to $V^{(d)}$, in Section V.

B. Convergence of Mutual Information for General DMCs

Given a general DMC W with input alphabet $\mathcal{X}$ (not necessarily binary) and output alphabet $\mathcal{Y}$, consider the d-view DMC $W^{(d)}$. Let the input to $W^{(d)}$ be X, with input distribution being P_X, and let its outputs be $Y_1, \ldots, Y_d \in \mathcal{Y}^d$. The joint distribution of the inputs and outputs is such that $P_{X,Y^d}(x, y^d) = P_X(x) \prod_{i=1}^d W(y_i|x)$. Much like the proof of Corollary IV.1, the inequality

$$\mathsf{H}(X|Y^d) \leq Z_g(W)^d,$$

can be shown to hold (see [24, Prop. 4.8]), where $Z_g(W) := \sum_{x\neq x'} \sum_{y\in\mathcal{Y}} \sqrt{P(x)W(y|x)P(x')W(y|x')}$ is a scaled version of the Bhattacharya parameter for general DMCs. This then results in $\rho(I) \geq -\log Z_g(W)$, thereby showing that for well-behaved DMCs, the rate of convergence of $I^{(d)}$ to $H(X)$ is exponential in d. We shall now take up the question of the exact speed of this convergence.

To this end, we will be next interested in evaluating, up to exponential tightness, the behaviour of

$$\begin{aligned}\mathsf{H}(X|Y^d) &= \mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(X \mid Y^d)}\right] \\ &= \sum_{x\in\mathcal{X}} P_X(x)\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right]. \end{aligned} \tag{5}$$

Now, fix an $x \in \mathcal{X}$ and focus on the inner term in (5). Then,

$$\begin{aligned}&\mathbb{E}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right] \\ &\qquad = \int_0^\infty \mathbb{P}\left[\log \frac{1}{P_{X|Y^d}(x \mid Y^d)} \geq t\middle| X = x\right] \mathrm{d}t. \end{aligned} \tag{6}$$

The inner probability, to wit, $p_x(t) := \mathbb{P}\left[-\log P_{X|Y^d}(x \mid Y^d) \geq t | X = x\right]$ is bounded as follows:

Lemma IV.3. *We have that*

$$\frac{p_x(t)}{(|\mathcal{X}|-1)} \leq \max_{\tilde{x}\neq x} \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-(t-\log(|\mathcal{X}|-1))}}{1-e^{-t}}\middle| X = x\right],$$

and

$$p_x(t) \geq \max_{\tilde{x}\neq x} \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right].$$

Proof. First, observe that

$$\begin{aligned} &p_x(t) \\ &= \mathbb{P}\left[\frac{P_{XY^d}(x, Y^d)}{P_{Y^d}(Y^d)} \leq e^{-t}\middle| X = x\right] \\ &= \mathbb{P}\left[P(x)P(Y^d \mid x) \leq e^{-t} \sum_{\tilde{x}} P(\tilde{x})P(Y^d \mid \tilde{x})\middle| X = x\right] \\ &= \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{\sum_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right]. \end{aligned}$$

We first prove the upper bound in the lemma. The above probability obeys

$$\begin{aligned}
&\mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{\sum_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\
&\leq \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{(|\mathcal{X}|-1)\max_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\
&= \mathbb{P}\left[\frac{1}{|\mathcal{X}|-1}\cdot\min_{\tilde{x}\neq x}\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\
&\leq \sum_{\tilde{x}\neq x}\mathbb{P}\left[\frac{1}{|\mathcal{X}|-1}\cdot\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right] \\
&\leq (|\mathcal{X}|-1)\cdot\max_{\tilde{x}\neq x}\mathbb{P}\left[\frac{1}{|\mathcal{X}|-1}\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}}\middle| X = x\right],
\end{aligned}$$

thereby showing the upper bound. Further, for any $\tilde{x}$, we have that

$$\begin{aligned}
p_x(t) &= \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{\sum_{\tilde{x}\neq x} P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}} \mid X = x\right] \\
&\geq \max_{\tilde{x}\neq x}\mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-t}}{1-e^{-t}} \mid X = x\right],
\end{aligned}$$

thereby proving the lower bound. □

Since we are interested mainly in the behaviour of $\mathsf{H}(X|Y^d)$ upto first order in the exponent d, it suffices for us to understand the behaviour of $p_x(t)$ upto first order in the exponent d with the expectation that we will accrue an additional $\Theta(\log d)$ term in the exponent, in the final expression (see Eq. (6) above and [25, Sec. 4.2] for intuition). To this end, following Lemma IV.3, for fixed $z \in [0,\infty)$ and $c > 0$, we shall focus on the term

$$\Gamma_{x,\tilde{x}}(z) := \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{ce^{-z}}{1-e^{-z}}\middle| X = x\right],$$

with the reasoning that the rate of convergence of $\mathsf{H}(X|Y^d)$ will be dictated by the smallest rate of $\Gamma_{x,\tilde{x}}(z)$, as z increases from 0 to ∞.

Before we proceed, we require some more notation. Let us denote the log-likelihood ratio by

$$L^y_{x,\tilde{x}} := \log\frac{P_{Y|X}(y \mid x)}{P_{Y|X}(y \mid \tilde{x})}.$$

For a given $y^d \in \mathcal{Y}^d$ and for $b \in \mathcal{Y}$, let $\hat{Q}_{y^d}(b) := \frac{1}{d}\sum_{i=1}^d \mathbb{1}\{y_i = b\}$ be the empirical frequency of the letter b in any output sequence. Further, we define

$$L_{x,\tilde{x}}(\hat{Q}) := \sum_{b\in\mathcal{Y}}\hat{Q}(b)L^b_{x,\tilde{x}}.$$

Finally, let

$$v_{x,\tilde{x}} \equiv v_{x,\tilde{x}}(z) := \frac{1}{d}\left(\log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(\tilde{x})}\right)$$

and let $z(v_{x,\tilde{x}}) := \log\left(1 + \frac{cP(\tilde{x})}{P(x)}\cdot e^{-dv_{x,\tilde{x}}}\right)$ be its inverse function.

Proposition IV.1. *We have that for any fixed $\delta > 0$,*

1) *If $v_{x,\tilde{x}} \geq L_{x,\tilde{x}}(P_{Y|x}) + \delta$, then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = \Theta(1)$.*
2) *If $v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|x}) - \delta$, then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = e^{-dE(v_{x,\tilde{x}})+\Theta(\log d)}$, where*

$$E(v_{x,\tilde{x}}) := \begin{cases} \max_{\lambda\geq 0}\left[-\log\left(\sum_y P(y|x)^{1-\lambda}P(y|x')^{\lambda}\right) - \lambda v_{x,\tilde{x}}\right], \\ \qquad\qquad \text{if } v_{x,\tilde{x}} \geq \min_{b\in\mathcal{Y}} L^b_{x,\tilde{x}}, \\ \infty, \text{ otherwise.} \end{cases}$$

Proof. Observe that for any $z \in [0,\infty)$,

$$\begin{aligned}
&\Gamma_{x,\tilde{x}}(z) \\
&= \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{ce^{-z}}{1-e^{-z}}\middle| X = x\right] \\
&= \mathbb{P}\left[\sum_{i=1}^d \log\frac{P_{Y|X}(Y_i \mid x)}{P_{Y|X}(Y_i \mid \tilde{x})} \leq \log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(\tilde{x})}\middle| X = x\right] \\
&= \mathbb{P}\left[\sum_{i=1}^d L^{Y_i}_{x,\tilde{x}} \leq \log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(\tilde{x})}\middle| X = x\right] \\
&= \mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L^{Y^d}_{x,\tilde{x}}\right] \leq v_{x,\tilde{x}}(z)\middle| X = x\right].
\end{aligned}$$

Writing this differently, we have that for any $v_{x,\tilde{x}} \in \mathbb{R}$,

$$\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = \mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L^{Y^d}_{x,\tilde{x}}\right] \leq v_{x,\tilde{x}}\middle| X = x\right].$$

Consider first the case when $v_{x,\tilde{x}} \geq L_{x,\tilde{x}}(P_{Y|x}) + \delta$. Note that $L_{x,\tilde{x}}(P_{Y|x}) = D(P_{Y|x}||P_{Y|\tilde{x}}) > 0$. We split this case into two subcases.

- If $v_{x,\tilde{x}} \geq \max_{b\in\mathcal{Y}} L^b_{x,\tilde{x}}$ then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = 1$.
- Otherwise, consider the case where $L_{x,\tilde{x}}(P_{Y|x}) + \delta \leq v_{x,\tilde{x}} \leq \max_{b\in\mathcal{Y}} L^b_{x,\tilde{x}}$. Now, observe that

$$\begin{aligned}
&1 \geq \Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) \geq \\
&\qquad \mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L^{Y^d}_{x,\tilde{x}}\right] \leq L_{x,\tilde{x}}(P_{Y|x}) + \delta\middle| X = x\right].
\end{aligned}$$

  Furthermore, by a simple application of Markov's inequality, we get that since $L_{x,\tilde{x}}(P_{Y|x}) = \mathbb{E}_{Y^d\sim P_{Y^d|x}}[L^{Y^d}_{x,\tilde{x}}]$, the following inequality holds:

$$\mathbb{P}\left[\mathbb{E}_{\hat{Q}}\left[L^{Y^d}_{x,\tilde{x}}\right] \leq L_{x,\tilde{x}}(P_{Y|x}) + \delta\middle| X = x\right] \geq \frac{\delta}{L_{x,\tilde{x}}(P_{Y|x}) + \delta}.$$

  Hence, we have that $1 \geq \Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) \geq \frac{\delta}{L_{x,\tilde{x}}(P_{Y|x})+\delta}$.

Hence, overall, if $v_{x,\tilde{x}} \geq L_{x,\tilde{x}}(P_{Y|x}) + \delta$, we get that $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = \Theta(1)$.

Next, consider the case when $v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|x}) - \delta$. We again split this case into two subcases.

- If $v_{x,\tilde{x}} \leq \min_{b\in\mathcal{Y}} L^b_{x,\tilde{x}}$, then $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = 0$.
- If $\min_{b\in\mathcal{Y}} L^b_{x,\tilde{x}} \leq v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|X=x})$, then this is a large-deviations event. The probability $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}}))$ thus decays exponentially in d, i.e., $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}})) = e^{-dE_1(v_{x,\tilde{x}})+\Theta(\log d)}$, for some $E_1 : \mathbb{R} \to [0,\infty)$, where by Sanov's theorem (see [15, Sec. 11.4]), we have that $E_1(v_{x,\tilde{x}}) = \min_{Q:\, L_{x,\tilde{x}}(Q)\leq v_{x,\tilde{x}}} D(Q \mid\mid P_{Y|X=x})$. We now aim to characterize this function E_1.
  From convex duality (see [28, Sec. 5.2]), we obtain that

$$E_1(v_{x,\tilde{x}}) = \max_{\lambda\geq 0}\min_{Q} D(Q \mid\mid P_{Y|X=x}) + \lambda(L_{x,\tilde{x}}(Q) - v_{x,\tilde{x}}). \tag{7}$$

First, let us solve the inner minimization. Fix $\lambda \geq 0$. Observe that the term $f(\lambda, Q) := D(Q \mid\mid P_{Y|X=x}) + \lambda L_{x,\tilde{x}}(Q)$ obeys

$$\begin{aligned} f(\lambda, Q) &= \sum_y Q(y) \left(\log \frac{Q(y)}{P_{Y|X=x}(y)} + \lambda L_{x,\tilde{x}}(y) \right) \\ &= -\sum_y Q(y) \left(\log \frac{P_{Y|X=x}(y)}{Q(y) e^{\lambda L_{x,\tilde{x}}(y)}} \right) \\ &\geq -\log \sum_y \frac{P_{Y|X=x}(y)}{e^{\lambda L_{x,\tilde{x}}(y)}}, \end{aligned}$$

using Jensen's inequality. Also note that equality holds when $Q(y) = Q^*(y) = \frac{1}{Z_{x,\tilde{x}}(\lambda)} P_{Y|X=x}(y) e^{-\lambda L_{x,\tilde{x}}(y)}$, where $Z_{x,\tilde{x}}(\lambda) := \sum_{y \in \mathcal{Y}} P_{Y|X=x}(y) e^{-\lambda L_{x,\tilde{x}}(y)}$ is the partition function (normalization constant). Substituting Q^* back into the function results

$$\begin{aligned} f(\lambda, Q^*) &= -\log Z_{x,\tilde{x}}(\lambda) - \lambda v_{x,\tilde{x}} \\ &= -\log \left(\sum_y P_{Y|X=x}(y) e^{-\lambda L_{x,\tilde{x}}(y)} \right) - \lambda v_{x,\tilde{x}} \\ &= -\log \left(\sum_y P_{Y|X=x}^{1-\lambda}(y \mid x) P_{Y|X}^{\lambda}(y \mid \tilde{x}) \right) - \lambda v_{x,\tilde{x}}. \end{aligned}$$

Plugging back into (7), we get that

$$\begin{aligned} &E_1(v_{x,\tilde{x}}) \\ &= \max_{\lambda \geq 0} \left[-\log \left(\sum_y P_{Y|X=x}^{1-\lambda}(y \mid x) P_{Y|X}^{\lambda}(y \mid \tilde{x}) \right) - \lambda v_{x,\tilde{x}} \right]. \end{aligned}$$

Note that by setting $E(v_{x,\tilde{x}}) = E_1(v_{x,\tilde{x}})$ when $\min_{b \in \mathcal{Y}} L_{x,\tilde{x}}^b \leq v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|X=x})$ and $E(v_{x,\tilde{x}}) = \infty$ when $v_{x,\tilde{x}} \leq \min_{b \in \mathcal{Y}} L_{x,\tilde{x}}^b$, we obtain a tight characterization of the rate of exponential decay of $\Gamma_{x,\tilde{x}}(z(v_{x,\tilde{x}}))$, in these ranges of $v_{x,\tilde{x}}$. □

We now state a prove a lemma that will allow us to identify the exact rate of exponential decay of $\mathsf{H}(X|Y^d)$.

Lemma IV.4. *We have that*

$$\int_0^\infty \max_{\tilde{x} \neq x} \Gamma_{x,\tilde{x}}(z) \mathrm{d}z = e^{\Theta(\log d) + \Theta(\log |\mathcal{X}|)} \cdot e^{-d \cdot \min_{\tilde{x} \neq x} \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}})}.$$

Proof. Consider an integral of the form

$$\bar{I} = \int_0^\infty \max_{\tilde{x} \neq x} \Gamma_{x,\tilde{x}}(z) \mathrm{d}z.$$

First, observe that

$$\begin{aligned} \bar{I} &\leq \int_0^\infty \sum_{\tilde{x} \neq x} \Gamma_{x,\tilde{x}}(z) \mathrm{d}z \\ &\leq |\mathcal{X}|^2 \cdot \max_{\tilde{x} \neq x} \int_0^\infty \Gamma_{x,\tilde{x}}(z) \mathrm{d}z \\ &= e^{\Theta(\log |\mathcal{X}|)} \cdot \max_{\tilde{x} \neq x} \int_0^\infty \Gamma_{x,\tilde{x}}(z) \mathrm{d}z. \end{aligned}$$

Further,

$$\bar{I} \geq \max_{\tilde{x} \neq x} \int_0^\infty \Gamma_{x,\tilde{x}}(z) \mathrm{d}z.$$

Hence, overall, we have that $\bar{I} = \exp(\Theta(\log |\mathcal{X}|)) \cdot \max_{\tilde{x} \neq x} \int_0^\infty \Gamma_{x,\tilde{x}}(z) \mathrm{d}z$. We next focus on the integral

$$I = \int_0^\infty \Gamma_{x,\tilde{x}}(z) \mathrm{d}z,$$

for a fixed $x, \tilde{x}$.

By the definition $v_{x,\tilde{x}} = v_{x,\tilde{x}}(z) = \frac{1}{d} \left(\log \left(\frac{ce^{-z}}{1-e^{-z}} \right) - \log \frac{P(x)}{P(x')} \right)$ and from Prop. IV.1, we see that the integral I can be split into two integrals for its evaluation: integral I_1, when $z \leq \log \left(1 + \frac{cP(\tilde{x})}{P(x)} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta)} \right) := u(d)$ (or equivalently, $v_{x,\tilde{x}} \geq L_{x,\tilde{x}}(P_{Y|x}) + \delta$), and integral I_2, when $z \geq \log \left(1 + \frac{cP(\tilde{x})}{P(x)} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x}) - \delta)} \right) := \ell(d)$ (or equivalently, $v_{x,\tilde{x}} \leq L_{x,\tilde{x}}(P_{Y|x}) - \delta$), with $I = I_1 + I_2$. Here, we pick $\delta < L_{x,\tilde{x}}(P_{Y|x})$.

Consider first the integral I_1. We have that

$$\begin{aligned} I_1 &= \int_{z \leq u(d)} \Gamma_{x,\tilde{x}}(z) \mathrm{d}z \\ &= \Theta(1) \cdot u(d). \end{aligned}$$

Now, we claim that $u(d) = e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta) + \Theta(\log d)}$. To see this, let $\tilde{c} := \frac{cP(\tilde{x})}{P(x)}$ and observe that from standard identities involving the log function,

$$u(d) \leq \tilde{c} e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta)},$$

and

$$u(d) \geq \frac{\tilde{c} e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta)}}{1 + \tilde{c} e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta)}} \geq \frac{\tilde{c}}{2} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta)},$$

since $L_{x,\tilde{x}}(P_{Y|x}) = D(P_{Y|x} || P_{Y|\tilde{x}}) > 0$. Hence, overall, we have that for any fixed $\delta > 0$, $I_1 = e^{-d(L_{x,\tilde{x}}(P_{Y|x}) + \delta) + \Theta(1)}$. Taking the limit as $\delta \downarrow 0$, we get that

$$I_1 = e^{-dL_{x,\tilde{x}}(P_{Y|x}) + \Theta(1)} = e^{-d \cdot D(P_{Y|x} || P_{Y|\tilde{x}}) + \Theta(1)}. \tag{8}$$

Consider now the integral I_2. Note that from case 2) in Prop. IV.1, when $z \geq \log \left(1 + \tilde{c} e^{-d \cdot \min_b L_{x,\tilde{x}}^b} \right) := U(d)$, we have that $I_2 = 0$. Hence, we can write

$$I_2 = \int_{\ell(d) \leq z \leq U(d)} \Gamma_{x,\tilde{x}}(z) \mathrm{d}z,$$

where, for this interval $\ell(d) \leq z \leq U(d)$ of interest,

$$\Gamma_{x,\tilde{x}}(z) = e^{-dE(v_{x,\tilde{x}}(z)) + \Theta(\log d)},$$

with

$$E(v_{x,\tilde{x}}(z)) = \max_{\lambda \geq 0} \left[-\log \left(\sum_y P(y|x)^{1-\lambda} P(y|x')^{\lambda} \right) - \lambda v_{x,\tilde{x}} \right], \tag{9}$$

from Prop. IV.1.

It is clear that $E(v_{x,\tilde{x}}(z))$ increases as $v_{x,\tilde{x}}$ decreases. In the integral I_2, we thus make the change of variables

$$v = \frac{1}{d} \left(\log \left(\frac{ce^{-z}}{1-e^{-z}} \right) - \log \frac{P(x)}{P(x')} \right),$$

with $\mathrm{d}z = \frac{-dce^{-dv}}{1+ce^{-dv}}\mathrm{d}v$. Thus,

$$\begin{aligned} I_2 &= dc \int_{v=\min_b L^b_{x,\tilde{x}}}^{L_{x,\tilde{x}}(P_{Y|x})-\delta} e^{-dE(v)} \cdot \frac{e^{-dv}}{1+ce^{-dv}}\mathrm{d}v \\ &= dc \int_{v=0}^{L_{x,\tilde{x}}(P_{Y|x})-\delta} e^{-dE(v)} \cdot \frac{e^{-dv}}{1+ce^{-dv}}\mathrm{d}v + \\ &\qquad dc \int_{v=\min_b L^b_{x,\tilde{x}}}^{0} e^{-dE(v)} \cdot \frac{e^{-dv}}{1+ce^{-dv}}\mathrm{d}v. \end{aligned}$$

Let us call the first integral in the expansion above as $I_{2,a}$ and the second as $I_{2,b}$ (note that $\min_b L^b_{x,\tilde{x}} \le 0$). Now, observe that

$$\begin{aligned} I_{2,a} &= dc \int_{v=0}^{L_{x,\tilde{x}}(P_{Y|x})-\delta} e^{-dE(v)} \cdot e^{-dv+\Theta(\log d)} dv \\ &= \exp(-d(E(L_{x,\tilde{x}}(P_{Y|x})) + L_{x,\tilde{x}}(P_{Y|x}) - \delta) + \Theta(\log d)). \end{aligned}$$

Here, in the first eqauility, we argue as previously that

$$\frac{e^{-dv}}{1+ce^{-dv}} = e^{-dv+\Theta(1)}.$$

Furthermore, the second equation follows from Laplace's method (e.g., see [25, Sec. 4.2]), using the fact that both $e^{-dE(v)}$ and e^{-dv} increase as v increases. Taking $\delta \downarrow 0$, we obtain that

$$I_{2,a} = e^{-d(E(L_{x,\tilde{x}}(P_{Y|x}))+L_{x,\tilde{x}}(P_{Y|x}))+\Theta(\log d)}. \tag{10}$$

Moreover,

$$\begin{aligned} I_{2,b} &= e^{\Theta(\log d)} \cdot \int_{v=\min_b L^b_{x,\tilde{x}}}^{0} e^{-dE(v)}\mathrm{d}v \\ &= e^{-dE(0)+\Theta(\log d)} = e^{-d\cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}. \end{aligned} \tag{11}$$

Here, for the first equality, we use the fact that when $v \le 0$,

$$\frac{1}{2} \le \frac{e^{-dv}}{1+ce^{-dv}} \le \frac{1}{c},$$

and hence $\frac{e^{-dv}}{1+ce^{-dv}} = \Theta(1)$. The second equality holds again by Laplace's method, using the fact that $e^{-dE(v)}$ increases with v.

Putting everything together, in order to nail down the exponential behaviour of integral I, we need to reconcile Equations (8), (10), and (11), to understand which among I_1, $I_{2,a}$, and $I_{2,b}$, is the largest. First, observe that since $\mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) \le D(P_{Y|x}||P_{Y|\tilde{x}})$, we have that $I_{2,b}$ dominates I_1.

Therefore, the exponential behaviour of integral I is governed by the rate

$$\overline{E} = \min\left\{E(L_{x,\tilde{x}}(P_{Y|x})) + L_{x,\tilde{x}}(P_{Y|x}), \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}})\right\}.$$

Now note that the maximum over $\lambda \ge 0$ in the expressions for E and C can be rewritten as a maximum over $\lambda \in [0,1]$ (owing to the concavity of $-\log\left(\sum_y P(y|x)^{1-\lambda}P(y|x')^{\lambda}\right)$ and the fact that it equals 0 at $\lambda = 0, 1$). We then get that

$$E(L_{x,\tilde{x}}(P_{Y|x})) + L_{x,\tilde{x}}(P_{Y|x}) \ge \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}),$$

implying that $\overline{E} = \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}})$. Therefore, $I = e^{-d\cdot\mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$, and hence that $\overline{I} = \max_{\tilde{x}\neq x} \exp(-d\cdot \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|))$. □

We are now ready to prove Theorem III.1 for the convergence of the mutual information.

Proof of Thm. III.1 for mutual information. Recall from Lemma IV.3 that

$$p_x(t) \le (|\mathcal{X}|-1)\cdot \max_{\tilde{x}\neq x} \Gamma_{x,\tilde{x}}(t),$$

with the constant c in the definition of $\Gamma_{x,\tilde{x}}(t)$ taken to be $|\mathcal{X}|-1$. Thus, from Lemma IV.4 and (6), we get that

$$\begin{aligned} &\mathbb{E}\left[\log\frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right] \\ &= \int_0^\infty p_x(t)\mathrm{d}t \\ &\le \exp\left(-d\cdot\min_{\tilde{x}\neq x}\mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right), \end{aligned}$$

where the inequality holds by the Laplace method [25, Sec. 4.2]. Similarly, since

$$p_x(t) \ge \max_{\tilde{x}\neq x} \Gamma_{x,\tilde{x}}(t),$$

with the constant c in the definition of $\Gamma_{x,\tilde{x}}(t)$ taken to be 1, we get that

$$\begin{aligned} &\mathbb{E}\left[\log\frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right] \\ &\ge \exp\left(-d\cdot\min_{\tilde{x}\neq x}\mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right). \end{aligned}$$

Thus,

$$\begin{aligned} &\mathbb{E}\left[\log\frac{1}{P_{X|Y^d}(x \mid Y^d)}\middle| X = x\right] \\ &= \exp\left(-d\cdot\min_{\tilde{x}\neq x}\mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right). \end{aligned}$$

Plugging back in (5), we obtain that $\mathsf{H}(X|Y^d) = \max_{x,\tilde{x}:\tilde{x}\neq x} \exp\left(-d\cdot\mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right)$, yielding the required result. □

V. Convergence of Channel Dispersion of DMCs

We now shift our attention to proving the convergence of the dispersion $V^{(d)} = \mathbb{E}\left[\left(\iota(X;Y^d) - I(X;Y^d)\right)^2\right]$, to the varentropy $\mathsf{V}(X)$ of the input distribution P_X, for general DMCs W.

First, we write

$$\begin{aligned} &V^{(d)} \\ &= \mathbb{E}\left[\left(\log\frac{P(X|Y^d)}{P(X)} - \left(\mathsf{H}(X) - \mathsf{H}(X|Y^d)\right)\right)^2\right] \\ &= \mathbb{E}\left[\left(\log\frac{1}{P(X)} - H(X)\right)^2\right] + \mathbb{E}\left[\left(\log P(X|Y^d) + \mathsf{H}(X|Y^d)\right)^2\right] \\ &\quad + 2\cdot\mathbb{E}\left[(\log P(X|Y^d) + \mathsf{H}(X|Y^d))\cdot\left(\log\frac{1}{P(X)} - H(X)\right)\right] \\ &= \mathsf{V}(X) + \left(\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] - \mathsf{H}(X|Y^d)^2\right) + \theta_d, \end{aligned} \tag{12}$$

where we use the fact that $\mathbb{E}\left[-\log P(X|Y^d)\right] = \mathsf{H}(X|Y^d)$, and the cross-term θ_d is

$$2\cdot\mathbb{E}\left[(\log P(X|Y^d) + \mathsf{H}(X|Y^d))\cdot\left(\log\frac{1}{P(X)} - H(X)\right)\right].$$

Further, observe that when $P_X = \mathrm{Unif}(\mathcal{X})$, the cross term vanishes, since for any $x \in \mathcal{X}$, we have that $\log \frac{1}{P(x)} = \log |\mathcal{X}| = \mathsf{H}(X)$. Our proof strategy hence is to first characterize (upto exponential tightness) the behaviour of $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right]$ and then handle the cross-term θ_d.

Lemma V.1. *We have that*

$$\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] = e^{\Theta(\log d)+\Theta(\log|\mathcal{X}|)} \cdot \max_{x,\tilde{x}:\tilde{x}\neq x} e^{-d\cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})}.$$

Proof. This proof is entirely analogous to the proof of convergence of $I^{(d)}$ to $\mathsf{H}(X)$, for general DMCs, in Section IV-B, and we only highlight the key steps.

We first consider the term $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right]$. As in (5), we first expand this term as:

$$\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] = \sum_{x\in\mathcal{X}} P(x)\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2 \middle| X = x\right],$$

where for any $x \in \mathcal{X}$, we have

$$\begin{aligned}
&\mathbb{E}\left[\left(\log P(x|Y^d)\right)^2 \middle| X = x\right] \\
&= \int_{t=0}^{\infty} \Pr\left[\left(\log P(x|Y^d)\right)^2 \geq t \middle| X = x\right] \mathrm{d}t \\
&= \int_{t=0}^{\infty} \mathbb{P}\left[\log P(x|Y^d) \geq \sqrt{t} \middle| X = x\right] \mathrm{d}t + \\
&\qquad \int_{t=0}^{\infty} \mathbb{P}\left[\log P(x|Y^d) \leq -\sqrt{t} \middle| X = x\right] \mathrm{d}t \\
&= \int_{t=0}^{\infty} \mathbb{P}\left[\log P(x|Y^d) \leq -\sqrt{t} \middle| X = x\right] \mathrm{d}t.
\end{aligned}$$

Here, the last equality holds since $P(x|Y^d) \leq 1$, for any $Y^d \in \mathcal{Y}^d$. As earlier, our task reduces to understanding the exponential behaviour of $q_x(t) := \Pr\left[\log P(x|Y^d) \leq -\sqrt{t} \middle| X = x\right]$.

Exactly as in Lemma IV.3, we have that

$$\begin{aligned}
&\frac{q_x(t)}{(|\mathcal{X}|-1)} \\
&\leq \max_{\tilde{x}\neq x} \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-(\sqrt{t}-\log(|\mathcal{X}|-1))}}{1-e^{-\sqrt{t}}} \middle| X = x\right],
\end{aligned}$$

and

$$q_x(t) \geq \max_{\tilde{x}\neq x} \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{e^{-\sqrt{t}}}{1-e^{-\sqrt{t}}} \middle| X = x\right].$$

Consider, for a fixed $c > 0$ and $z \in [0,\infty)$, the expression

$$\Delta_{x,\tilde{x}}(z) := \mathbb{P}\left[\frac{P(x)P(Y^d \mid x)}{P(\tilde{x})P(Y^d \mid \tilde{x})} \leq \frac{ce^{-\sqrt{z}}}{1-e^{-\sqrt{z}}} \middle| X = x\right].$$

As argued in the case of the mutual information, the rate of exponential decay of $\Delta_{x,\tilde{x}}(z)$, as a function of z, dictates the rate of exponential decay of $q_x(t)$, and hence of $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right]$. Once again, letting $v_{x,\tilde{x}} = v_{x,\tilde{x}}(z) = \frac{1}{d}\left(\log\left(\frac{ce^{-z}}{1-e^{-z}}\right) - \log\frac{P(x)}{P(x')}\right)$ and setting $(z(v_{x,\tilde{x}}))^{1/2} := \log\left(1 + \frac{cP(\tilde{x})}{P(x)} e^{-dv_{x,\tilde{x}}}\right)$, we obtain a result similar to Proposition IV.1, with the only modification being the replacement of $\Gamma_{x,\tilde{x}}$ by $\Delta_{x,\tilde{x}}$.

Next, as in Lemma IV.4, consider the integral

$$\overline{I}^V = \int_0^{\infty} \max_{\tilde{x}\neq x} \Delta_{x,\tilde{x}}(z)\mathrm{d}z.$$

We claim that $\overline{I}^V = e^{\Theta(\log d)+\Theta(\log|\mathcal{X}|)} \cdot e^{-d\min_{\tilde{x}\neq x} \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})}$. To obtain this, we argue as in the proof of Lemma IV.4 that $\overline{I}^V = e^{\Theta(\log|\mathcal{X}|)} \cdot \max_{\tilde{x}\neq x} I^V$, where

$$I^V = \int_0^{\infty} \Delta_{x,\tilde{x}}(z)\mathrm{d}z,$$

for a fixed $x, \tilde{x}$.

Now, we split the evaluation of I^V into two integrals: integral I_1^V, when $z \leq \left(\log\left(1 + \frac{cP(\tilde{x})}{P(x)} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})+\delta)}\right)\right)^2$, and integral I_2^V, when $z \geq \left(\log\left(1 + \frac{cP(\tilde{x})}{P(x)} \cdot e^{-d(L_{x,\tilde{x}}(P_{Y|x})-\delta)}\right)\right)^2$, with $I^V = I_1^V + I_2^V$. By arguments analogous to those in the proof of Lemma IV.4, we obtain that

$$I_1^V = e^{-2d\cdot D(P_{Y|x}||P_{Y|\tilde{x}})+\Theta(\log d)},$$

and

$$I_2^V = e^{-d\cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}.$$

Thus, we have that $I^V = e^{-d\cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})+\Theta(\log d)}$, and hence that $\overline{I}^V = \exp\left(-d \cdot \min_{\tilde{x}\neq x} \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right)$. Putting everything togehter, we get that $\mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] = \max_{x,\tilde{x}:\tilde{x}\neq x} \exp\left(-d \cdot \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right)$. □

Lemma V.2. *We have that*

$$\theta_d = e^{\Theta(\log d)+\Theta(\log|\mathcal{X}|)} \cdot \max_{x,\tilde{x}:\tilde{x}\neq x} e^{-d\cdot \mathsf{C}(P_{Y|x},P_{Y|\tilde{x}})}.$$

Proof. Recall that the cross term

$$\theta_d := 2 \cdot \mathbb{E}\left[\left(\log P(X|Y^d) + \mathsf{H}(X|Y^d)\right) \cdot \left(\log\frac{1}{P(X)} - H(X)\right)\right].$$

We then write

$$\begin{aligned}
&\theta_d(x) \\
&:= \left(\log\frac{1}{P(x)} - H(X)\right) \cdot \left(\mathbb{E}\left[\log P(x|Y^d) \middle| X = x\right] + \mathsf{H}(X|Y^d)\right),
\end{aligned}$$

with $\theta_d = \sum_{x\in\mathcal{X}} P(x)\theta_d(x)$. From the capacity analysis, we know that (see (6))

$$\begin{aligned}
&\mathbb{E}\left[\log P(x|Y^d) \middle| X = x\right] \\
&= -\max_{x'\neq x} \exp\left(-d \cdot \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right).
\end{aligned}$$

From our previous analysis, we know that $\mathsf{H}(X|Y^d) = \max_{x,x':x'\neq x} \exp\left(-d \cdot \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right)$. Hence,

$$\begin{aligned}
\theta_d &= e^{\Theta(\log d)} \cdot \theta_d(x) \\
&= \max_{x,x':x\neq x'} \exp\left(-d \cdot \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right).
\end{aligned}$$

□

With these lemmas in place, the proof of Theorem is easily completed.

Proof of Thm. III.1 for dispersion. From (12) we see that

$$V^{(d)} - \mathsf{V}(X) = \mathbb{E}\left[\left(\log P(X|Y^d)\right)^2\right] - \mathsf{H}(X|Y^d)^2 + \theta_d.$$

Putting together Lemmas V.1 and V.2 and the fact that $\mathsf{H}(X|Y^d) = \max_{x,x':x'\neq x} \exp\left(-d\cdot \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}}) + \Theta(\log d) + \Theta(\log|\mathcal{X}|)\right)$ (from the proof of Thm. III.1 for mutual information), we readily obtain the required result. □

VI. Convergence for Multi-Letter Channels and Some Examples

Recall that Theorem III.1 presented the convergence of mutual information and channel dispersion for d-view DMCs $W^{(d)}$, where the channel W is specified by *single-letter* transition probabilities $\{W(y|x) : x \in \mathcal{X}, y \in \mathcal{Y}\}$. In this section, we first extend the statement of this theorem and its proof to multi-letter channels. Consider a multi-letter channel $W_{n,\nu}$, for any $n, \nu \geq 0$ that are independent of d, specified by a channel law $\{W(y^\nu|x^n) = P_{Y^\nu|X^n}(y^\nu|x^n) : x^n \in \mathcal{X}^n, y^\nu \in \mathcal{Y}^\nu\}$. Here, the input alphabet is $\mathcal{X}^n$ and the output alphabet is $\mathcal{Y}^\nu$. We work with the d-view multi-letter channel $W^{(d)}_{n,\nu}$ defined by the channel law

$$P(Y^{d\nu} = y^{d\nu} \mid X^n = x^n) = \prod_{i=1}^{d} W_{n,\nu}(y^\nu|x^n).$$

Recall here our assumption that $|\mathcal{X}|$ and $|\mathcal{Y}|$ do not depend on d. Similar to the earlier definition, we let

$$I^{(d)}_{n,\nu} := I(X^n; Y^{d\nu}) \tag{13}$$

to be the mutual information rate over $W^{(d)}_{n,\nu}$. Here, we assume as before that the input distribution P_{X^n} is arbitrary, but fixed; in particular, it is d-independent . Likewise, let the channel dispersion be

$$V^{(d)}_{n,\nu} := \mathbb{E}\left[\left(\iota(X^n; Y^{d\nu}) - I(X^n; Y^{d\nu})\right)^2\right], \tag{14}$$

where $\iota(x; y^d) := \log \frac{P(y^{d\nu}|x^n)}{P(y^{d\nu})}$. The following corollary of Theorem III.1 holds:

Corollary VI.1. *We have that*

$$I^{(d)}_{n,\nu} = H(X^n) - \exp\left(-d\rho_{n,\nu} + \Theta(\log d) + n\cdot\Theta(\log|\mathcal{X}|)\right), \text{ and}$$
$$V^{(d)}_{n,\nu} = \exp\left(-d\rho_{n,\nu} + \Theta(\log d) + n\cdot\Theta(\log|\mathcal{X}|)\right),$$

where $\rho_{n,\nu} = \min_{x^n,\tilde{x}^n:x^n\neq\tilde{x}^n} \mathsf{C}(P_{Y^\nu|x^n}, P_{Y^\nu|\tilde{x}^n})$.

Note that the $n\cdot\Theta(\log|\mathcal{X}|)$ terms above arise due to the fact that the input alphabet size is now $|\mathcal{X}|^n$, instead of $|\mathcal{X}|$, as in Theorem III.1.

Now, consider the special case when $\nu = n$ and the channel $W_{n,n}$ corresponds to an n-letter DMC; more precisely, suppose that the multi-letter channel law obeys

$$W_{n,n}(y^n|x^n) = \prod_{i=1}^{n} W(y_i|x_i).$$

Now, for the *single-letter* channel W as above, let $\rho = \min_{x,\tilde{x}:x\neq\tilde{x}} \mathsf{C}(P_{Y|x}, P_{Y|\tilde{x}})$. We then have the following corollary:

Corollary VI.2. *For an n-letter DMC* $W_{n,n}$,

$$I^{(d)}_{n,n} = H(X^n) - \exp\left(-dn\rho + \Theta(\log d) + n\cdot\Theta(\log|\mathcal{X}|)\right), \text{ and}$$
$$V^{(d)}_{n,n} = \exp\left(-dn\rho + \Theta(\log d) + n\cdot\Theta(\log|\mathcal{X}|)\right).$$

Proof. The proof will follow from Corollary VI.1 if we can show that $\rho_{n,n} = n\rho$, for an n-letter DMC $W_{n,n}$. To this end, let $u, v \in \mathcal{X}$ be such that $\rho = \mathsf{C}(P_{Y|u}, P_{Y|v})$. Further, observe that for any two inputs $x^n \neq \tilde{x}^n$, we have $\mathsf{C}(P_{Y^n|x^n}, P_{Y^n|\tilde{x}^n}) = \sum_{i=1}^n \mathsf{C}(P_{Y|x_i}, P_{Y|\tilde{x}_i})$. Hence,

$$\begin{aligned}\rho_{n,n} &= \min_{x^n,\tilde{x}^n:x^n\neq\tilde{x}^n} \sum_{i=1}^{n} \mathsf{C}(P_{Y|x_i}, P_{Y|\tilde{x}_i}) \\ &\geq \sum_{i=1}^{n} \min_{x_i,\tilde{x}_i:x_i\neq\tilde{x}_i} \mathsf{C}(P_{Y|x_i}, P_{Y|\tilde{x}_i}) = n\rho,\end{aligned}$$

where equality in the second step is obtained when $x^n = (u, u, \ldots, u)$ and $\tilde{x}^n = (v, v, \ldots, v)$. □

While Corollary VI.1 treats n and ν as fixed constants, one can generalize the definitions of the d-view mutual information and channel dispersion to the setting where ν is a *random variable* too. In particular, let $I^{(d)}_n$ and $V^{(d)}_n$ be defined the same way as in (13) and (14), respectively, but where the expectations are taken with respect to the randomness in the length ν of the outputs too. Furthermore, let

$$f_\lambda(x^n, \tilde{x}^n) := \sum_{m\geq 0} \Pr[\nu = m] \sum_{y^m} P^{1-\lambda}(y^m|x^n) P^\lambda(y^m|\tilde{x}^n).$$

We then define $\rho_n := \min_{x^n\neq\tilde{x}^n} \max_{\lambda\in[0,1]} -\log f_\lambda(x^n, \tilde{x}^n)$.

In what follows, we present some estimates of ρ_n for the deletion channel, where the length $\nu \leq n$ of the outputs is random.

A. Upper Bounds on ρ_n for the Deletion Channel

In this subsection, we consider the multi-letter deletion channel, which is an important example of a synchronization channel. Recall that for $\delta \in (0, 1)$, the deletion channel $\mathrm{Del}(\delta)$ deletes each input bit $x_i \in \mathcal{X}$, $1 \leq i \leq n$, with probability δ.

For $\mathrm{Del}(\delta)$, for any fixed $x^n, \tilde{x}^n$ with $x^n \neq \tilde{x}^n$, and for any fixed $\lambda \in (0, 1)$, we have

$$\begin{aligned}&f_\lambda(x^n, \tilde{x}^n) \\ &= \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \sum_{y^m} N(x^n \to y^m)^{1-\lambda} \cdot N(\tilde{x}^n \to y^m)^\lambda,\end{aligned} \tag{15}$$

where $N(u^n \to v^m)$, for $u^n \in \{0, 1\}^n$, $v^m \in \{0, 1\}^m$, is the number of occurrences of v^m as a (potentially non-contiguous) subsequence in u^n.

In what follows, we present three upper bounds on ρ_n for $\mathrm{Del}(\delta)$. These bounds are obtained by fixing specific sequences $x^n, \tilde{x}^n \in \mathcal{X}^n$ and using $\max_{\lambda\in[0,1]} -\log f_\lambda(x^n, \tilde{x}^n)$, evaluated at these chosen sequences, as an upper bound on ρ_n.

Lemma VI.1. *For Del(δ), we have that* $\rho_n \leq -n\log\delta$.

Proof. Consider *any* fixed $x^n, \tilde{x}^n$ with $x^n \neq \tilde{x}^n$. For any $\lambda \in [0,1]$, $f_\lambda(x^n, \tilde{x}^n) \geq \delta^n$, from (15), since the empty string occurs exactly once as a subsequence of both x^n and $\tilde{x}^n$. Hence, we have that $\rho_n = \min_{x^n \neq \tilde{x}^n} \max_{\lambda \in [0,1]} -\log f_\lambda(x^n, \tilde{x}^n) \leq -n \log \delta$. □

From the above lemma and from Corollary VI.2, we see that the rate ρ_n for the deletion channel is *smaller* than the rate $\rho_{n,n}$ for the n-letter erasure channel with erasure probability δ. Indeed, recall that for the BEC(δ), we have that $\rho = -\log \delta$, and hence, by Corollary VI.2, $\rho_{n,n} = -n \log \delta$.

The bound that follows improves on this naïve upper bound, for a certain range of δ values, and for large enough n. Recall that $h_b(\cdot)$ denotes the binary entropy function. For simplicity, in the next lemma, we assume that n is even; the case of odd n can be handled with minor modifications to the exisiting proof.

Lemma VI.2. *For Del(δ), we have that*

$$\rho_n \leq \left(n \cdot \left(-\log(1-\delta) - \frac{3h_b(1/3)}{2} \right) + O(\log n) \right)^+,$$

where $(a)^+$ *denotes* $\max\{a, 0\}$*, for* $a \in \mathbb{R}$.

Proof. We pick sequences $x^n, \tilde{x}^n$, each of which has the largest number of subsequences (see [29, Sec. III]), and hence also the largest number of common subsequences. In particular, we pick $x^n, \tilde{x}^n$ to be, respectively, the alternating sequences $01010\ldots$ and $10101\ldots$; each having length n. The intuition behind choosing these sequences is to try to obtain a small upper bound on ρ_n by seeking to make $f_\lambda(x^n, \tilde{x}^n)$ large.

Now, for any $y^m \in \{0,1\}^m$, with $m \leq n$, we see that $N(x^n \to y^m) = N(\tilde{x}^n \to y^m)$. Furthermore, $N(x^n \to y^m) > 0$ iff $w(y^m) \in [m - n/2, n/2]$; we then call y^m "admissible". In this case, we have that

$$N(x^n \to y^m) = \binom{n/2}{w(y^m)} \cdot \binom{n/2}{m - w(y^m)} = N(\tilde{x}^n \to y^m).$$

In the above, we implicitly assume that the binomial coefficient $\binom{k}{t} = 0$ if $t < 0$ or $t > k$. Thus, for $\lambda \in (0,1)$,

$$\begin{aligned} & f_\lambda(x^n, \tilde{x}^n) \\ &= \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \sum_{y^m \text{ admissible}} \binom{n/2}{w(y^m)} \cdot \binom{n/2}{m - w(y^m)} \\ &= \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \sum_{w=\max\{m-n/2,0\}}^{\min\{m,n/2\}} \binom{m}{w} \cdot \binom{n/2}{w} \cdot \binom{n/2}{m-w} \\ &=: \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \cdot \sum_{w=0}^{\min\{m,n/2\}} \binom{m}{w} \cdot \binom{n/2}{w} \cdot \binom{n/2}{m-w}. \end{aligned} \tag{16}$$

Let $\alpha_1 := \sum_{w=0}^{m} \binom{m}{w} \cdot \binom{n/2}{w} \cdot \binom{n/2}{m-w}$ and let $\alpha_2 := \sum_{w=0}^{n/2} \binom{m}{w} \cdot \binom{n/2}{w} \cdot \binom{n/2}{m-w}$. Consider α_1, for the case when $m \leq n/2$. We then have that

$$\begin{aligned} \alpha_1 &\geq \min_{0 \leq \omega \leq m} \binom{n/2}{m-\omega} \cdot \sum_{w=0}^{m} \binom{m}{w} \cdot \binom{n/2}{w} \\ &= \binom{n/2}{0} \cdot \binom{n/2+m}{m} = \binom{n/2+m}{m}, \end{aligned}$$

where the first equality follows from the Chu-Vandermonde identity (see, e.g., [30, p. 42]). Likewise, consider α_2, for the case when $m \geq n/2$. Here,

$$\begin{aligned} \alpha_2 &= \sum_{w=0}^{n/2} \binom{m}{w} \cdot \binom{n/2}{n/2-w} \cdot \binom{n/2}{m-w} \\ &\geq \min_{0 \leq \omega \leq n/2} \binom{n/2}{m-\omega} \cdot \sum_{w=0}^{n/2} \binom{m}{w} \cdot \binom{n/2}{n/2-w} \\ &= \binom{n/2}{0} \cdot \binom{n/2+m}{m} = \binom{n/2+m}{m}, \end{aligned}$$

where the second equality above again uses the Chu-Vandermonde identity. Hence, plugging our observations back in (16), we obtain that

$$\begin{aligned} f_\lambda(x^n, \tilde{x}^n) &\geq \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \cdot \binom{n/2+m}{m} \\ &\geq (1-\delta)^n \cdot \binom{3n/2}{n/2} \\ &\geq (1-\delta)^n \cdot \exp\left(\frac{3n}{2} \cdot h_b(1/3) - O(\log n) \right), \end{aligned}$$

where the last (asymptotic) inequality follows from [15, Thm. 11.1.3]. Furthermore, we have that $f_\lambda(x^n, \tilde{x}^n) = 1$, for $\lambda \in \{0,1\}$. The upper bound in the lemma then follows, readily. □

Note that the expression $g(\delta) := -\log(1-\delta) - \frac{3h_b(1/3)}{2}$ is in fact non-positive for $\delta \lessapprox 0.6151$. Hence, in this range of δ values, for large enough n, we have that $\rho_n = 0$, for Del(δ), and hence that $I_n^{(d)}$ and $V_n^{(d)}$ converge only polynomially quickly (in d) to their respective limits.

In the next lemma, we show that for all $\delta \in (0,1)$ (and hence for $\delta \gtrapprox 0.6151$ too), the rate of convergence ρ_n grows much slower than linear in n.

Lemma VI.3. *For Del(δ), we have that*

$$\rho_n \leq \max_{\lambda \in [0,1]} \lambda \log n - \log \mathbb{E}_{L \sim Bin(n,\delta)} \left[L^\lambda \right].$$

Proof. As before, we pick specific sequences $x^n, \tilde{x}^n$ and seek to upper bound $\rho_{n,\nu}$ by $\max_{\lambda \in [0,1]} -\log f_\lambda(x^n, \tilde{x}^n)$, for this choice of $x^n, \tilde{x}^n$. Here, we choose $x^n = 0^n$ and $\tilde{x}^n = 0^{n-1}1$.

Clearly, for any $m \leq n$, we have that $y^m \in \{0,1\}^n$ is a subsequence of x^n iff $y^m = 0^m$. For $y^m = 0^m$, hence, it follows that

$$N(x^n \to y^m) = \binom{n}{m} \text{ and } N(\tilde{x}^n \to y^m) = \binom{n-1}{m}.$$

Thus, for $\lambda \in (0,1)$,

$$\begin{aligned} f_\lambda(x^n, \tilde{x}^n) &= \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \cdot \binom{n}{m}^{1-\lambda} \cdot \binom{n-1}{m}^{\lambda} \\ &= \sum_{m=0}^{n} \delta^{n-m}(1-\delta)^m \cdot \binom{n}{m} \cdot \left(\frac{n-m}{n} \right)^{\lambda} \\ &= \mathbb{E}_{L \sim \text{Bin}(n,1-\delta)} \left[\left(1 - \frac{L}{n} \right)^{\lambda} \right] \\ &= \frac{1}{n^\lambda} \cdot \mathbb{E}_{L \sim \text{Bin}(n,\delta)} \left[L^\lambda \right], \end{aligned} \tag{17}$$

and the claim follows by noting that $f_\lambda(x^n, \tilde{x}^n) = 1$, for $\lambda \in \{0, 1\}$. □

While it is possible to obtain asymptotic (in n) expressions for the fractional moment $\mathbb{E}\left[L^\lambda\right]$ above, for $\lambda \in [0, 1]$, we instead seek to obtain a simple upper bound on the expression in Lemma VI.3, which holds for all n. We thus obtain the following corollary:

Corollary VI.3. *For Del(δ), we have that $\rho_n \leq \log n$.*

Proof. Observe that for $L \sim \mathrm{Bin}(n, \delta)$, we have that $\max_{\lambda \in [0,1]} -\log \mathbb{E}_{L \sim \mathrm{Bin}(n,\delta)}\left[L^\lambda\right] = 0$, attained when $\lambda = 0$. Hence,

$$\begin{aligned}\rho_n &\leq \max_{\lambda \in [0,1]} \lambda \log n - \log \mathbb{E}_{L \sim \mathrm{Bin}(n,\delta)}\left[L^\lambda\right] \\ &\leq \max_{\lambda \in [0,1]} \lambda \log n + \max_{\lambda \in [0,1]} \left(-\log \mathbb{E}_{L \sim \mathrm{Bin}(n,\delta)}\left[L^\lambda\right]\right) \\ &= \log n.\end{aligned}$$

The corollary thus follows. □

Hence, putting together Lemma VI.2, and Corollary VI.3, we see that for the deletion channel Del(δ), the rate $\rho_n = 0$ for $\delta \lessapprox 0.6151$ and for large enough n. Furthermore, for all n and for all $\delta \in (0, 1)$, ρ_n grows much slower than linear in n, unlike the quantity $\rho_{n,n}$ corresponding to n-letter DMCs (see Corollary VI.2); in fact, ρ_n grows at most logarithmically in n.

VII. Poisson Approximation Channel

In this section, we take a detour from the asymptotic regime of large d considered in the rest of the paper, and seek to obtain tight, non-asymptotic bounds on the capacity of a special multi-view DMC, which holds for all values of d. In particular, we consider the special case of the d-view BSC(p), denoted by $\mathrm{BSC}^{(d)}(p)$. Recall that the capacity of this channel equals the capacity of the binomial channel $\mathsf{Bin}_d(p)$, which is given in (2). Our objective is to show that the Poisson approximation channel $\mathsf{Poi}_d(p)$ introduced in Section III has capacity that is a very good lower bound on the capacity of $\mathrm{BSC}^{(d)}(p)$, via the proof of Theorem III.3. Before we proceed with the proof, we state and prove a simple lemma about the concavity of the mutual information $I^{(d)} = I(X; Y^d)$, for any d-view DMC $W^{(d)}$ and input distribution P_X.

Lemma VII.1. *For any d-view DMC $W^{(d)}$ and input distribution P_X, we have that $I^{(d)}$ is concave over the positive integers, i.e.,*

$$I^{(d)} - I^{(d-1)} \geq I^{(d+1)} - I^{(d)},$$

for all $d \geq 1$.

Proof. We first rewrite what we intend to prove differently: since $I^{(m)} = H(X) + H(Y^m) - H(X, Y^m)$, for any $m \geq 0$, with $I^{(0)} := 0$, we would like to show that

$$\begin{aligned}&H(Y^d) - H(Y^{d-1}) + H(X, Y^{d-1}) - H(X, Y^d) \\ &\qquad \geq H(Y^{d+1}) - H(Y^d) + H(X, Y^d) - H(X, Y^{d+1}). \quad (18)\end{aligned}$$

However, observe that for any $m \geq 0$, $H(X, Y^{m+1}) - H(X, Y^m) = H(Y_{m+1}|X, Y^m) = H(Y_{m+1}|X)$, by the channel law of $W^{(d)}$. Hence, (18) reduces to showing that

$$H(Y^d) - H(Y^{d-1}) \geq H(Y^{d+1}) - H(Y^d). \quad (19)$$

Now, by the submodularity of entropy [22, Thm. 1.6], we have that

$$H(Y^{d-1}, Y_{d+1}) - H(Y^{d-1}) \geq H(Y^{d+1}) - H(Y^d).$$

The above implies (19), since the distributions of (Y^{d-1}, Y_{d+1}) and Y^d are equal, by the stationarity of the DMC $W^{(d)}$. □

For clarity, we first write down the capacity $C(\mathsf{Poi}_d(p))$ of the Poisson approximation channel. As argued in Section III, the capacity is achieved when the inputs $X \sim P_X = \mathrm{Ber}(1/2)$, yielding

$$\begin{aligned}&C(\mathsf{Poi}_d(p)) \\ &= \mathsf{H}(X) - \mathsf{H}(X|R_1, R_2) \\ &= 1 + \sum_{\substack{r_1 \geq 0, \\ r_2 \geq 0}} \frac{e^{-dp}(dp)^{r_1}}{r_1!} \cdot \frac{e^{-d(1-p)}(d(1-p))^{r_2}}{r_2!} \times \\ &\qquad \log \frac{p^{r_1}(1-p)^{r_2}}{p^{r_1}(1-p)^{r_2} + p^{r_2}(1-p)^{r_1}}. \quad (20)\end{aligned}$$

We now prove Theorem III.3.

Proof. Consider the mutual information between the input and outputs of the $\mathsf{Poi}_d(p)$ channel when P_X is uniform over the inputs. Since the number of outputs $N = R_1 + R_2$ is known to the decoder, we have that

$$\begin{aligned}C(\mathsf{Poi}_d(p)) &= I(X; R_1, R_2) \\ &= I(X; R_1, R_2 \mid N) \\ &= \sum_{n=0}^{\infty} \mathbb{P}[N = n] \cdot I(X; R_1, R_2 \mid N = n). \quad (21)\end{aligned}$$

Now, observe that since the sum of a $\mathrm{Poi}(dp)$ random variable and a $\mathrm{Poi}(d(1-p))$ random variable is distributed according to $\mathrm{Poi}(d)$, we have by symmetry that $N \sim \mathrm{Poi}(d)$. Further, it is easy to check that the conditional distribution $P_{R_1,R_2|N=n}$ obeys $P_{R_1,R_2|N=n}(r_1, r_2 \mid n) = \binom{n}{r_1} p^{r_1}(1-p)^{n-r_1}$, for all $r_1, r_2 \geq 0$ such that $r_1 + r_2 = n$. Hence, we have that $I(X; R_1, R_2 \mid N = n) = C(\mathsf{Bin}_n(p))$.

Thus, substituting in (21), we get that

$$C(\mathsf{Poi}_d(p)) = \mathbb{E}_{N \sim \mathrm{Poi}(d)}\left[C(\mathsf{Bin}_N(p))\right]. \quad (22)$$

Next, using the above pleasing expression, we show that $C(\mathsf{Poi}_d(p))$ is indeed a lower bound on $C(\mathsf{Bin}_d(p))$. From Lemma VII.1 we have that $C(\mathsf{Bin}_n(p))$ is concave over the positive integers. Therefore, applying Jensen's inequality, we get that

$$\begin{aligned}C(\mathsf{Poi}_d(p)) &= \mathbb{E}_{N \sim \mathrm{Poi}(d)}\left[C(\mathsf{Bin}_N(p))\right] \\ &\leq C(\mathsf{Bin}_{\mathbb{E}N}(p)) = C(\mathsf{Bin}_d(p)).\end{aligned}$$

Furthermore, we have that

$$\begin{aligned}
&C(\mathsf{Bin}_d(p)) - C(\mathsf{Poi}_d(p)) \\
&= C(\mathsf{Bin}_d(p)) - \mathbb{E}_{N\sim\mathrm{Poi}(d)}\left[C(\mathsf{Bin}_N(p))\right] \\
&\le C(\mathsf{Bin}_d(p)) - 1 + \mathbb{E}_{N\sim\mathrm{Poi}(d)}[Z(p)^N] \\
&\le \exp(-d(1-Z(p))) - Z(p)^{2d},
\end{aligned}$$

where the two inequalities make use of Corollary IV.1 and the last inequality uses the well-known expression for the moment-generating function of a Poisson random variable. □

Remark. For the binomial channel $\overline{\mathsf{Bin}}_n$, whose input is a real number in $[0,1]$, with n output trials, that is considered in [31] too, a suitable Poisson approximation channel $\overline{\mathsf{Poi}}_n$ can be designed. This channel takes as input $X \in [0,1]$ and outputs the pair (R_1, R_2) such that $P_{R_1|X} = \mathrm{Poi}(nX)$ and $P_{R_2|X} = \mathrm{Poi}(n(1-X))$. For $\overline{\mathsf{Poi}}_n$, as above, we will have that $C(\overline{\mathsf{Poi}}_n) = \mathbb{E}_{N\sim\mathrm{Poi}(n)}\left[C(\overline{\mathsf{Bin}}_N)\right]$. However, the proof of the concavity of the capacity $C(\overline{\mathsf{Bin}}_n)$ as in Lemma VII.1 will not hold, owing to the failure of the conditional independence assumption that is crucially employed in the proof. We are hence unable to obtain a lower bound on $C(\overline{\mathsf{Bin}}_n)$ via $C(\overline{\mathsf{Poi}}_n)$.

VIII. Conclusion and Future Work

In this paper, we considered the behaviour of the information rate and channel dispersion of multi-view noisy channels, with arbitrary, but fixed, input distributions, in the regime where the number of views is very large. In the first part of the paper, we considered multi-view discrete memoryless channels (DMCs), and showed that the information rate and the channel dispersion converge exponentially quickly to the input entropy and varentropy, respectively, with an input distribution-independent rate that equals the smallest Chernoff information between two conditional distributions of the output given different inputs. We then extended these results to general multi-letter channels, and computed explicit upper bounds on the Chernoff information above for the deletion channel. Finally, for the special case of the multi-view binary symmetric channel, we also presented a close, non-asymptotic approximation to its capacity, via the capacity of the so-called Poisson approximation channel.

One interesting direction for future work would be the extension of the techniques and results in this paper to other multi-view channels with memory and synchronization channels (see [32] and [33, Sec. VII]).

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References

[1] R. Heckel, G. Mikutis, and R. N. Grass, "A characterization of the DNA data storage channel," *Scientific Reports*, vol. 9, 2018. [Online]. Available: https://api.semanticscholar.org/CorpusID:4743189

[2] I. Shomorony and R. Heckel, "Information-theoretic foundations of DNA data storage," *Foundations and Trends® in Communications and Information Theory*, vol. 19, no. 1, pp. 1–106, 2022. [Online]. Available: http://dx.doi.org/10.1561/0100000117

[3] N. Weinberger, "Error probability bounds for coded-index DNA storage systems," *IEEE Transactions on Information Theory*, vol. 68, no. 11, pp. 7005–7022, 2022.

[4] N. Weinberger and N. Merhav, "The DNA storage channel: Capacity and error probability bounds," *IEEE Transactions on Information Theory*, vol. 68, no. 9, pp. 5657–5700, 2022.

[5] M. Kovačević and V. Y. F. Tan, "Codes in the space of multisets—coding for permutation channels with impairments," *IEEE Transactions on Information Theory*, vol. 64, no. 7, pp. 5156–5169, 2018.

[6] A. Lenz, P. H. Siegel, A. Wachter-Zeh, and E. Yaakobi, "Coding over sets for DNA storage," *IEEE Transactions on Information Theory*, vol. 66, no. 4, pp. 2331–2351, 2020.

[7] V. Levenshtein, "Efficient reconstruction of sequences," *IEEE Transactions on Information Theory*, vol. 47, no. 1, pp. 2–22, 2001.

[8] R. Gabrys, H. M. Kiah, and O. Milenkovic, "Asymmetric Lee distance codes: New bounds and constructions," in *2015 IEEE Information Theory Workshop (ITW)*, 2015, pp. 1–5.

[9] M. Mitzenmacher, "On the theory and practice of data recovery with multiple versions," in *2006 IEEE International Symposium on Information Theory*, 2006, pp. 982–986.

[10] I. Land and J. Huber, "Information combining," *Foundations and Trends® in Communications and Information Theory*, vol. 3, no. 3, pp. 227–330, 2006. [Online]. Available: http://dx.doi.org/10.1561/0100000013

[11] I. Sutskover, S. Shamai, and J. Ziv, "Extremes of information combining," *IEEE Transactions on Information Theory*, vol. 51, no. 4, pp. 1313–1325, 2005.

[12] T. Batu, S. Kannan, S. Khanna, and A. McGregor, "Reconstructing strings from random traces," in *Proceedings of the Fifteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, ser. SODA '04. USA: Society for Industrial and Applied Mathematics, 2004, p. 910–918.

[13] M. Cheraghchi, R. Gabrys, O. Milenkovic, and J. Ribeiro, "Coded trace reconstruction," *IEEE Transactions on Information Theory*, vol. 66, no. 10, pp. 6084–6103, 2020.

[14] Y. Peres and A. Zhai, "Average-case reconstruction for the deletion channel: Subpolynomially many traces suffice," in *2017 IEEE 58th Annual Symposium on Foundations of Computer Science (FOCS)*, 2017, pp. 228–239.

[15] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-India, 2010.

[16] C. E. Shannon, "A mathematical theory of communication," *Bell Systems Technical Journal*, vol. 27, no. 3, pp. 379–423, July 1948.

[17] R. G. Gallager, *Stochastic Processes: Theory for Applications*. Cambridge: Cambridge University Press, 2013.

[18] T. Richardson and R. Urbanke, *Modern Coding Theory*. Cambridge University Press, 2008.

[19] A. Guillen i Fabregas, I. Land, and A. Martinez, "Extremes of error exponents," *IEEE Transactions on Information Theory*, vol. 59, no. 4, pp. 2201–2207, 2013.

[20] Y. Polyanskiy, H. V. Poor, and S. Verdu, "Channel coding rate in the finite blocklength regime," *IEEE Transactions on Information Theory*, vol. 56, no. 5, pp. 2307–2359, 2010.

[21] V. Y. F. Tan, "Asymptotic estimates in information theory with non-vanishing error probabilities," *Foundations and Trends® in Communications and Information Theory*, vol. 11, no. 1-2, pp. 1–184, 2014. [Online]. Available: http://dx.doi.org/10.1561/0100000086

[22] Y. Polyanskiy and Y. Wu, *Information Theory: From Coding to Learning*. Cambridge University Press, 2023+ (*book draft*).

[23] I. Land, S. Huettinger, P. Hoeher, and J. Huber, "Bounds on information combining," *IEEE Transactions on Information Theory*, vol. 51, no. 2, pp. 612–619, 2005.

[24] E. Şaşoğlu, "Polarization and polar codes," *Foundations and Trends® in Communications and Information Theory*, vol. 8, no. 4, pp. 259–381, 2012. [Online]. Available: http://dx.doi.org/10.1561/0100000041

[25] N. Merhav, "Statistical physics and information theory," *Foundations and Trends® in Communications and Information Theory*, vol. 6, no. 1–2, pp. 1–212, 2010. [Online]. Available: http://dx.doi.org/10.1561/0100000052

[26] Z. Zhou and P. Li, "Non-asymptotic Chernoff lower bound and its application to community detection in stochastic block model," *arXiv e-prints*, p. arXiv:1812.11269, Dec. 2018.

[27] C. Shannon, R. Gallager, and E. Berlekamp, "Lower bounds to error probability for coding on discrete memoryless channels. I," *Information and Control*, vol. 10, no. 1, pp. 65–103, 1967. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0019995867900526

[28] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge: Cambridge University Press, 2004.

[29] A. Lenz, Y. Liu, C. Rashtchian, P. H. Siegel, A. Wachter-Zeh, and E. Yaakobi, "Coding for efficient dna synthesis," in *2020 IEEE International Symposium on Information Theory (ISIT)*, 2020, pp. 2885–2890.

[30] W. Koepf, *Hypergeometric Summation An Algorithmic Approach to Summation and Special Function Identities*. Vieweg, 1998.

[31] R. D. Wesel, E. E. Wesel, L. Vandenberghe, C. Komninakis, and M. Medard, "Efficient binomial channel capacity computation with an application to molecular communication," in *2018 Information Theory and Applications Workshop (ITA)*, 2018, pp. 1–5.

[32] B. Haeupler and M. Mitzenmacher, "Repeated deletion channels," in *2014 IEEE Information Theory Workshop (ITW 2014)*, 2014, pp. 152–156.

[33] M. Cheraghchi and J. Ribeiro, "An overview of capacity results for synchronization channels," *IEEE Transactions on Information Theory*, vol. 67, no. 6, pp. 3207–3232, 2021.